

Common-heat replica variance, raw sequential owners, and the post-heat boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-31 · this version issued 2026-07-31 · v1.0

1. Purpose and scope

R-135 left a seemingly small operator question between its proved uncentred R-088 atom estimate and its conditional variance summation: could one commute legal future conditioning through the spatial and root-OU operators and then centre the R-088 atom? The operator contraction is valid, but an index audit shows that this is not the production bridge. The literal R-088 atom has already been averaged over roots later than its current root. It is therefore measurable at the retained sigma-field and its future-centred part is zero. The missing issue is information loss, not a difficult commutator.

This note makes five exact advances and one route correction.

1. It proves the Hilbert conditional-centring contraction, records the exact random-operator commutator, and distinguishes current-root Doob centring from future-owner centring.
2. It proves that the post-heat R-088 sequential atom belongs to the retained sigma-field. Future-centring it is identically zero. A two-point fixture proves that post-heat mean data cannot recover future variance.
3. It puts the R-125 future variance into a common-heat two-replica form. The heat dummy is a common Hilbert coordinate; only the legal future is replicated.
4. It gives the exact pathwise raw sequential telescope in each replica and then subtracts the two telescopes. This is the correct algebraic owner intertwiner before future heat averaging. It keeps the complete low endpoint and recalculates all later feedback in each replica.
5. It rewrites the same variance by the original forward-filtration Doob increments and the root OU resolvent. The generic Hilbert identity and the regular R-102 cross-Doob transpose are prior authorities; they are reused and not advertised as a new R-136 theorem.
6. It records that full product-space Douglas–Loewner positivity is only a sufficient acceptance test on the actual production tangent graph. A graph-compressed direct Taylor estimate is a weaker live route and spends the source and sextic squares once. This observation does not repair the missing raw spatial estimate.

The scope is one finite product chart, one finite cutoff, one positive floor, one legal retained sigma-field per owner, deterministic R-088 target heat for the post-heat degeneracy, common heat in the replica Hilbert coordinate, and an orthogonal exhaustive output-cluster direct sum. Same-root visits are recombined before a variance, replica difference, or shell norm is formed. The result does not prove the production raw-current spatial atom bound, the once-owned q_k sum, the signed forest-current inequality, positive collar headroom, the complete-low anchor, either A13 gate, Nelson, or Sector A.

The binding authorities are R-079, R-084, R-088, R-099, R-102, R-104, R-119, R-123, R-125, R-127, R-128, R-129, R-132, and R-135. The new negative is a method no-go only. No prior result changes tier or scope.

2. Two orthogonal centring directions

Let

$$\mathcal{F}_{j-1} \subset \mathcal{F}_j \subset \mathcal{F}_J, \quad P_j X = \mathbb{E}[X \mid \mathcal{F}_j]. \quad (2.1)$$

There are two different projections adjacent to P_j :

$$d_j = P_j - P_{j-1}, \quad R_j = I - P_j. \quad (2.2)$$

The first extracts the current-root innovation. The second removes the retained data and extracts the entire future residual. Orthogonal projection algebra gives

$$d_j^2 = d_j, \quad R_j^2 = R_j, \quad d_j R_j = R_j d_j = 0. \quad (2.3)$$

R-084 applies the Gaussian OU resolvent to a d_j current-root innovation. R-125 defines its future variance by R_j after retaining the current root. These are orthogonal coordinates and cannot be renamed as each other.

Theorem 2.1 (conditional-centring contraction)

Let $P_{\mathcal{H}} = \mathbb{E}[\cdot \mid \mathcal{H}]$ on a Hilbert-valued L^2 space and put $C_{\mathcal{H}} = I - P_{\mathcal{H}}$. If L is a deterministic bounded contraction, or a bounded operator measurable with respect to $\mathcal{H}$, then

$$P_{\mathcal{H}}(LY) = LP_{\mathcal{H}}Y, \quad C_{\mathcal{H}}(LY) = LC_{\mathcal{H}}Y. \quad (2.4)$$

Consequently

$$\boxed{\mathbb{E}\|LC_{\mathcal{H}}Y\|^2 = \mathbb{E}\|LY\|^2 - \mathbb{E}\|P_{\mathcal{H}}(LY)\|^2 \leq \mathbb{E}\|LY\|^2.} \quad (2.5)$$

Proof. Bochner conditional expectation commutes with a deterministic bounded linear map. An $\mathcal{H}$ -measurable bounded multiplier may likewise be pulled out. Conditional expectation is the orthogonal projection onto $L^2(\mathcal{H}; \mathcal{K})$, so its range and kernel obey Pythagoras. This proves (2.4)–(2.5). $\square$

For a random operator Q for which every displayed product and conditional expectation is square-integrable (in particular, for essentially bounded Q), the exact defect is

$$\boxed{C_{\mathcal{H}}(QZ) - QC_{\mathcal{H}}Z = QP_{\mathcal{H}}Z - P_{\mathcal{H}}(QZ).} \quad (2.6)$$

Thus a future-dependent output multiplier need not commute with future conditioning and cannot be moved through it without an additional measurability or commutation hypothesis. A current-root-dependent multiplier may commute with future conditioning and still fail to commute with the current-root OU semigroup. Deterministic spatial Fourier projections are safe for both probability operations. In R-135 equation (5.1), the OU and output operations have already been applied before the variable is centred, so (2.5) needs no OU commutator at all.

The four-point audit makes the boundary exact. Let X, Y be independent Rademachers, retain X , put $Q = \mathbf{1}_{\{Y=1\}}$, and set $Z = Y$. Then

$$C_{\mathcal{H}}(QZ) = Y/2, \quad QC_{\mathcal{H}}Z = Q, \quad C_{\mathcal{H}}(QZ) - QC_{\mathcal{H}}Z = -1/2, \quad (2.7)$$

so the squared commutator norm is $1/4$.

3. The literal R-088 future-centred atom vanishes

Fix root j . R-088 uses

$$\Phi_{A,j} = P_{\Sigma_{j+1}:J} F_A, \quad w_j = U_{j-1} + g_j, \quad (3.1)$$

and predictable control shells $a_k, k \leq j$. In this section write $\Phi = \Phi_{A,j}$ and use the R-088 prefix convention $B_{j,k} = w_j + A^{(k-1)}$, where $A^{(k-1)}$ is the sum of the already inserted predictable controls before shell k . Its exact atom is the three-channel derivative

$$\begin{aligned} T_{A,j,k}[v] &= \{D\Phi(B_{j,k} + a_k) - D\Phi(B_{j,k})\}[v]^T DB_{j,k} \\ &\quad + D\Phi(B_{j,k} + a_k)[v]^T Da_k \\ &\quad + \{\Phi(B_{j,k} + a_k) - \Phi(B_{j,k})\}^T Dv. \end{aligned} \quad (3.2)$$

The future heat in (3.1) is already a deterministic coefficient function. w_j is $\mathcal{F}_j$ -measurable, and every $a_k, k \leq j$, is $\mathcal{F}_{j-1}$ -measurable. Therefore

$$T_{A,j,k} \in L^2(\mathcal{F}_j; \mathcal{K}). \quad (3.3)$$

Current-root OU evolution and deterministic spatial or output projection preserve $\mathcal{F}_j$ -measurability. If R-125's legal future expectation is $\mathbb{E}_f = P_j$, then

$$\boxed{\mathbb{E}_f T_{A,j,k} = T_{A,j,k}, \quad (\mathbf{I} - \mathbb{E}_f) \Pi_m P_t^{(j)} T_{A,j,k} = 0.} \quad (3.4)$$

Equation (3.4) does not falsify R-135 Theorem 5.1. That theorem is a correct conditional summation lemma for a genuinely future-random atom family. It shows instead that the literal post-heat R-088 atom is not that family. The remaining bridge is an information and owner-index problem, not the contraction (2.5).

Theorem 3.1 (post-heat mean-only information loss)

Let the retained sigma-field be trivial and let $Y = \pm 1$ with equal probability. Put

$$X_0 = 0, \quad X_1 = Y. \quad (3.5)$$

Then

$$\mathbb{E}_f X_0 = \mathbb{E}_f X_1 = 0, \quad \text{Var}_f(X_0) = 0, \quad \text{Var}_f(X_1) = 1. \quad (3.6)$$

Hence no rule depending only on the post-heat conditional mean, or on deterministic transforms of that mean, recovers future variance for every raw current.

This is NG-2026-07-31-A13-POSTHEAT-MEAN-ONLY-FUTURE-VARIANCE-RECOVERY. It does not refute a raw-current lift, a two-replica lift, a common-terminal Doob decomposition, the complete production action, or Sector A.

4. Common-heat replica completion of the R-125 owner

Fix one complete owner o after visits and exhaustive output clusters have been recombined. Let

$$X_o^f \in \mathcal{K}_o = L^2(\mu_\Sigma; \bigoplus_c \mathcal{H}_{o,c}) \quad (4.1)$$

be its unaveraged six-row current. The coordinate r of $L^2(\mu_\Sigma)$ is the common heat dummy. Let f, f' be conditionally independent copies of the legal future given the retained sigma-field. The heat dummy, retained variables, output family, and trace covariance are the same in both copies.

Theorem 4.1 (paired-replica variance and owner identity)

For every square-integrable owner current,

$$\begin{aligned} \|\mathbb{E}_f X_o^f\|^2 &= \mathbb{E}_{f,f'} \operatorname{Re} \langle X_o^f, X_o^{f'} \rangle, \\ \mathcal{V}_o &= \mathbb{E}_f \|X_o^f - \mathbb{E}_f X_o^f\|^2 \\ &= \frac{1}{2} \mathbb{E}_{f,f'} \|X_o^f - X_o^{f'}\|^2. \end{aligned} \tag{4.2}$$

If Θ_o is the once-owned primitive trace and

$$\mathbf{F}_{063,o}^{\text{ad}} = \mathbb{E}_f \|X_o^f\|^2 - \Theta_o, \tag{4.3}$$

then the R-123 direct owner is

$$\begin{aligned} \mathcal{P}_{\text{comp},o} &= \frac{1}{2} (\|\mathbb{E}_f X_o^f\|^2 - \Theta_o) \\ &= \frac{1}{2} \mathbf{F}_{063,o}^{\text{ad}} - \frac{1}{4} \mathbb{E}_{f,f'} \|X_o^f - X_o^{f'}\|^2. \end{aligned} \tag{4.4}$$

Proof. Conditional independence gives the first line of (4.2). Expanding the replica difference gives twice the conditional second moment minus twice the conditional mean square, which is twice the variance. Substitution into the R-125 identity $2\mathcal{P}_{\text{comp},o} = \mathbf{F}_{063,o}^{\text{ad}} - \mathcal{V}_o$ proves (4.4). $\square$

The factor 1/4 in (4.4) is the product of the owner half and the replica half. Common heat is load bearing. Independently replicating the heat dummy would add a heat variance that does not belong to the R-125 future owner. Freezing later feedback in one or both replicas would remove part of the raw current and is likewise illegal.

The exact vector fixture used by both executables has future samples

$$X^- = (0, 3), \quad X^+ = (2, 1). \tag{4.5}$$

Its mean-square, variance, and second moment are 5, 2, and 7. With $\Theta = 3$, the forest is 4 and both forms of (4.4) give $\mathcal{P}_{\text{comp}} = 1$.

5. The raw sequential replica telescope

The replica identity becomes useful only before future heat has discarded the raw path. Fix integers $k_0 < K$. For each replica $q \in \{f, f'\}$, let $X_o^q[A]$ denote the complete owner current evaluated with all legal field and control feedback of that replica. For $k_0 \leq k \leq K$, let

$$A^{(k),q} = A^{(k_0),q} + \sum_{k_0 < s \leq k} a_s^q. \tag{5.1}$$

Every future control is recomputed from the history of its own replica. Put

$$Y_{o,k_0}^q = X_o^q[A^{(k_0),q}], \quad Y_{o,k}^q = X_o^q[A^{(k),q}] - X_o^q[A^{(k-1),q}], \quad k_0 < k \leq K. \tag{5.2}$$

Theorem 5.1 (raw owner-to-sequential intertwiner)

At every finite cutoff, with $X_o^q := X_o^q[A^{(K),q}]$,

$$X_o^q = Y_{o,k_0}^q + \sum_{k_0 < k \leq K} Y_{o,k}^q, \tag{5.3}$$

and therefore

$$X_o^f - X_o^{f'} = (Y_{o,k_0}^f - Y_{o,k_0}^{f'}) + \sum_{k_0 < k \leq K} (Y_{o,k}^f - Y_{o,k}^{f'}). \quad (5.4)$$

For every deterministic bounded operator L applied identically in both replicas,

$$\text{Var}_f(LX_o) = \frac{1}{2} \mathbb{E}_{f,f'} \left\| L \left[(Y_{o,k_0}^f - Y_{o,k_0}^{f'}) + \sum_{k_0 < k \leq K} (Y_{o,k}^f - Y_{o,k}^{f'}) \right] \right\|^2. \quad (5.5)$$

Proof. Equation (5.3) is the pathwise endpoint telescope in a fixed replica. Subtract the two exact telescopes and then apply Theorem 4.1 to LX_o . No conditional-expectation commutator is used. $\square$

For one raw atom the same calculation gives

$$\frac{1}{2} \mathbb{E} \|L(Y_{o,k}^f - Y_{o,k}^{f'})\|^2 = \mathbb{E} \text{Var}_f(LY_{o,k}) \leq \mathbb{E} \|LY_{o,k}\|^2. \quad (5.6)$$

Thus a valid uncentred spatial estimate for the *raw complete atoms* would pass to their replica variance with no extra constant. R-135 proves the corresponding estimate only for the post-heat R-088 physical atom. Identifying the raw atoms in (5.2) with an R-088-type connection-complete family is the remaining production theorem, not a consequence of (5.6).

The complete-low atom in (5.4) may vanish only when it is genuinely the same in both futures. It cannot be dropped by naming it a baseline. Output clusters and visits must be recombined before (5.2); otherwise one replica variance is spuriously repeated over subowners.

6. Forward-filtration Doob coordinates and prior scope

For comparison, let $\tau(o)$ be the retained root of owner o and set

$$M_{o,r} = P_r X_o, \quad \Delta_{o,r} = M_{o,r} - M_{o,r-1}. \quad (6.1)$$

Hilbert martingale orthogonality gives

$$\mathbb{E}[\|X_o - M_{o,\tau(o)}\|^2 \mid \mathcal{F}_{\tau(o)}] = \sum_{r > \tau(o)} \mathbb{E}[\|\Delta_{o,r}\|^2 \mid \mathcal{F}_{\tau(o)}]. \quad (6.2)$$

After expectation and owner summation,

$$\sum_o \mathbb{E} \mathcal{V}_o = \sum_r \mathbb{E} \left\| \bigoplus_{\tau(o) < r} \Delta_{o,r} \right\|^2. \quad (6.3)$$

The direct sum in (6.3) is bookkeeping. It asserts no new physical orthogonality among different owners.

If root r is Gaussian and $M_{o,r}$ is in the root Sobolev domain, R-084's OU-resolvent identity gives, for deterministic Π_m ,

$$\mathbb{E} \|\Pi_m \Delta_{o,r}\|^2 = 2 \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(r)} D_r M_{o,r}\|_{\text{HS}}^2 dt. \quad (6.4)$$

Equations (6.2)–(6.4) use the original forward filtration. They are not a reverse-time reveal and do not change adapted controls.

R-079 already proves the fixed-terminal decomposition

$$f_j = d_j(\mathcal{J}_j - \mathcal{J}_0), \quad i_j = d_j(\mathcal{J}_* - \mathcal{J}_j), \quad f_j + i_j = d_j(\mathcal{J}_* - \mathcal{J}_0), \quad (6.5)$$

including both coefficient and derivative-feedback channels. R-102 already uses a cross-Doob finite-Fubini transpose for its narrower regular rational future-insertion current. R-136 therefore claims no novelty for Doob transposition itself. Its new connection is that the R-125 future variance and owner half factor admit the common-heat raw-replica telescope (4.4)–(5.5).

R-084’s predictable-root formula cannot be applied by substituting terminal A^* : that introduces feedback derivatives. R-099 gives an exact derivative-free Doob–Hardy ledger for the control coordinate $d_j A^*$, but does not control the complete nonlinear six-row multiplier. These authorities narrow the next task without closing it.

7. The future-feedback connection is load bearing

The smallest two-root energy fixture makes the missing channel quantitative. Let ξ_1, ξ_2 be independent centred unit-variance variables, put

$$\mathcal{J}_0 = 0, \quad \mathcal{J}_1 = \xi_1, \quad \mathcal{J}_* = 2\xi_1 + \xi_2. \quad (7.1)$$

The physical-prefix increments have energies 1 and 1, total 2. The fixed terminal has Doob energies 4 and 1, total 5. At the first root, $f_1 = \xi_1$ and $i_1 = \xi_1$, so the omitted square correction is

$$2\mathbb{E}\langle f_1, i_1 \rangle + \mathbb{E}\|i_1\|^2 = 2 + 1 = 3. \quad (7.2)$$

This reuses the filtration issue registered in NG-2026-07-30-A13-ROOTWISE-COMMON-TERMINAL-INFERENCE; it is not a new production counterexample.

A smooth Gaussian diagnostic also prevents deletion of the connection by formal predictability. For independent standard Gaussians ξ, η , let

$$h_n(\xi) = \frac{\sin(n\xi)}{n}, \quad X_n = \eta + h_n(\xi). \quad (7.3)$$

At the first reveal $M_1 = h_n(\xi)$ and $D_1 M_1 = \cos(n\xi)$. The exact OU-resolvent value is

$$\text{Var}(h_n) = \frac{1 - e^{-2n^2}}{2n^2} > 0. \quad (7.4)$$

Freezing later feedback would replace the nonzero connection by zero. This is a filtration diagnostic and an instance of the existing feedback derivative graph boundary, not an A1 action counterexample.

8. Minimal production one-use interface

The correct root index for a raw variance atom is its reveal root r , not only the starting owner $\tau(o)$. Suppose that the complete derivative in (6.4) is decomposed, after owner and cluster recombination, as

$$D_r M_{o,r} = D_r M_{o,r}^{\text{low}} + \sum_{A,k \leq r} \hat{T}_{o,A,r,k} + R_{o,r}^{\text{conn}}. \quad (8.1)$$

Every coefficient, derivative-feedback, six-row, rational, Cartan, trace, forest, heat, and low term must have exactly one owner. Define the reveal direct sum

$$\hat{\mathcal{T}}_{r,k}$$

only after all active owners and exhaustive clusters are placed inside it.

Fix $s = 2/3$ and $\gamma = 7/12$, and let $(q_k)_k$ be a deterministic nonnegative sequence. One sufficient spatial interface is

$$2 \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(r)} \hat{\mathcal{T}}_{r,k}\|_{\text{HS}}^2 dt \leq 2^{-2s(m-k)} q_k, \quad m \geq r + 5, \quad (8.2)$$

with

$$\sum_k q_k \leq Q. \quad (8.3)$$

The factor two in (8.2) is the OU factor and may be absorbed into q_k . Under (8.2)–(8.3), the R-135 nonorthogonal summation applies with its same geometric constant

$$C_{2/3,7/12}^{\text{var}} = 0.658881625872614\dots \quad (8.4)$$

No claim is made that (8.1)–(8.3) hold in production. The R-135 post-heat atom estimate does not prove them, and ownerwise $q_{o,k}$ costs may not be summed and then multiplied by an owner count.

The exact successor is therefore

$$\boxed{\text{A13-CLASSII-COMMON-HEAT-REPLICA-RAW-CURRENT-SEQUENTIAL-ONE-USE-BOUND.}} \quad (8.5)$$

It must first prove (8.1) with both R-079 feedback channels and the low atom, then prove (8.2)–(8.3), and only then test finite-collar headroom.

9. Production-graph compression and direct Taylor

R-127's three-channel Douglas matrix is necessary and sufficient when its source, sextic, and low variables range independently over the full product space. Production variations need not fill that product. At a fixed chart and base point, let

$$T_{t,h}u = (X_{t,h}u, Y_{t,h}u, L_{t,h}u). \quad (9.1)$$

The exact graph condition is

$$T_{t,h}^* \mathcal{M}_{t,h} T_{t,h} \succeq 0. \quad (9.2)$$

Full $\mathcal{M}_{t,h} \succeq 0$ implies (9.2), but the converse is false. The minimal scalar fixture

$$Tu = (u, u, u), \quad \mathcal{M} = \text{diag}(1, 1, -1) \quad (9.3)$$

has an ambient negative eigenvalue while

$$T^* \mathcal{M} T = 1 > 0. \quad (9.4)$$

This does not falsify R-127 or R-129; their universal product-space scope is correct. It prevents the full Douglas range and low-sign conditions from being treated as logically necessary for a direct production-graph proof.

R-119 and R-128 already supply the exact Taylor architecture. If a complete owner P obeys on every radial segment, where $\mathcal{S}_{\text{src}}(h) \geq 0$ and $\mathcal{S}_6(h) \geq 0$ are respectively the once-owned source and sextic quadratic functionals (not the components $X_{t,h}$ and $Y_{t,h}$ of (9.1)),

$$D^2 P(th)[h, h] \geq -2\eta \mathcal{S}_{\text{src}}(h) - 2\zeta \mathcal{S}_6(h) - 2C, \quad (9.5)$$

then

$$P(h) \geq P(0) + DP(0)[h] - \eta \mathcal{S}_{\text{src}}(h) - \zeta \mathcal{S}_6(h) - C. \quad (9.6)$$

If the absolute/linear anchor costs $\eta_0 \mathcal{S}_{\text{src}} + \zeta_0 \mathcal{S}_6 + C_0$, the action margins are

$$\frac{9}{20} - \eta - \eta_0, \quad \frac{3}{20} - \zeta - \zeta_0. \quad (9.7)$$

The half in (9.6) comes from $\int_0^1 (1-t) dt = 1/2$; the source and sextic squares are spent once. By contrast, full force completion costs 5/9 and consumes the displayed 9/20 source square before any second Hessian payment. Direct Taylor does not make $DP(0)$, the complete-low term, or the absolute anchor free.

The graph observation is a route clarification only. It cannot replace the raw owner construction in Sections 4–8.

10. Proof, failure, and successor map

Route	Verdict	Reusable conclusion
Conditional centring contraction	proved	Deterministic or retained-data operators contract future-centred energy; admissible random operators have the exact defect (2.6).
Post-heat R-088 future centring	failed method	The literal atom is $\mathcal{F}_j$ -measurable, so its R-125 future residual is zero.
Mean-only variance recovery	formal no-go	Equal post-heat means can have future variances zero and one.
Common-heat replica owner	proved	Equations (4.2)–(4.4) preserve the R-125 variance and owner half exactly.
Raw sequential telescope	proved braically	Equations (5.3)–(5.5) retain low and later feedback before future heat.
R-088 spatial transfer to raw atoms	open	R-135 controls the post-heat physical atom, not the connection-complete raw family.
Forward Doob transpose	reused	Generic Hilbert orthogonality and R-102 already own the transpose principle; R-136 uses it only as a coordinate.
Product Douglas necessity	rejected as necessity	Full product PSD is sufficient; the production graph may be positive with an indefinite ambient matrix.
Direct graph Taylor	active conditional route	It avoids source double spending but still needs the owner Hessian and absolute/low anchor.
Production one-use and headroom	open	Prove (8.1)–(8.3), then certify a finite collar and the complete low/matching/gap/anchor data.

The proof-search and audit records are EXP-000533–EXP-000539. They record the operator success, post-heat failure, replica/raw telescope, Doob scope audit, graph route, selected successor, and pre-release L^2 -admissibility correction separately so that none must be inferred from a later prose summary.

11. Devil’s-advocate review

Objection 1: Theorem 2.1 closes the R-135 owner intertwiner. UPHOLD against that inference. It closes only an operator contraction. Equation (3.4) shows that the literal post-heat atom has zero future-centred part and therefore carries no R-125 future variance.

Objection 2: Zero future residual of the post-heat atom proves the production future variance is zero. DISMISSED. The raw terminal current contains unaveraged future fields and later feedback. The two-point fixture (3.5)–(3.6) separates its variance from the identical post-heat mean.

Objection 3: Two replicas duplicate the heat and trace. VALID with mitigation. The heat dummy, retained variables, output family, and trace covariance are common. Only the legal future is replicated. Equation (4.4) uses the primitive trace once.

Objection 4: The future-Doob transpose is a new R-136 result. VALID correction. Generic martingale orthogonality is old, and R-102 already contains a production cross-Doob/Fubini transpose in its narrow regular rational class. R-136 claims only the new R-125 common-heat raw-replica owner connection.

Objection 5: The R-135 uncentred estimate transfers immediately to the raw atoms by (5.6). UPHELD against immediate transfer. Equation (5.6) is constant-free only after the same raw atom has an uncentred estimate. The R-135 atom has future heat already applied and omits the terminal feedback connection.

Objection 6: The physical R-088 atom and terminal atom differ only by harmless notation. DISMISSED. The two-root fixture has energies two and five; the missing cross-plus-square correction is exactly three. R-079 independently retains coefficient and derivative-feedback channels.

Objection 7: The graph fixture invalidates the Douglas criterion. DISMISSED. R-127's criterion is exact for all independent product variables. The fixture addresses necessity after restriction to $\text{Ran } T_{t,h}$ and creates no new negative authority.

Objection 8: Direct Taylor provides the missing absolute anchor. UPHELD against that inference. It avoids double spending but leaves $P(0) + DP(0)[h]$, the complete low block, and the absolute normalization to be proved and paid once.

Objection 9: R-136 closes controlled-shell one-use or Sector A. UPHELD against closure. No production proof of (8.1)–(8.3), signed forest-current bound, headroom, low/matching data, strict gap, anchor, A13 gate, Nelson estimate, or Sector-A endpoint is present.

12. Executable verification

Run from the repository root with the repository environment:

```
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
& $py '
  codes/foundations/a13_classii_common_heat_replica_raw_sequential_owner_boundary.py

& $py '
  codes/foundations/a13_classii_common_heat_replica_raw_sequential_owner_boundary_independent.py

& $py '
  codes/foundations/a13_classii_common_heat_replica_raw_sequential_owner_boundary_verify.py
```

Expected output is primary 71/71, non-importing standard-library independent 71/71, and integrated 225/225 PASS. The aggregate contract is 367/367. The independent route constructs the conditional-expectation matrices directly and does not import the primary implementation.

The executables check filtration projection algebra, conditional Pythagoras, the random-operator commutator, post-heat measurability, the mean-only no-go, paired-replica mean/variance/second-moment identities, the owner factor, the raw replica telescope, owner/reveal direct-sum accounting, the exact future-feedback energy gap, Gaussian OU chaos factors, graph compression, direct Taylor budgets, upstream manifests, source independence, and every scope firewall stated above.

The proof PDF is built from this exact source by

```
$note = 'claims/A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION/' +
  'notes/classii-common-heat-replica-raw-sequential-owner-boundary-' +
  '260731-v1.0.tex.txt'
& $py verification/scripts/build_note_pdf.py $note
```

Deterministic rebuild, extraction, security, form, nonblank-page, glyph, and all-page visual checks are pinned in the R-136 manifest.

13. Remaining theorem and no-overclaim statement

The next production theorem must:

1. construct the complete raw pre-heat atoms in (8.1), including the low atom, exhaustive clusters, all same-root visits, both R-079 feedback channels, all six rows, rational and Cartan companions, trace, forest, and common heat;
2. prove the projected raw atom estimate (8.2) without replacing the saturated current by the rejected extracted- Z^6 envelope;
3. prove the once-owned sum (8.3) without owner, reveal, visit, or cluster multiplicity;
4. compare the resulting far tail with independently certified finite near/balanced headroom;
5. prove the complete-low and matching-energy data, strict graph or Douglas gap, and absolute anchor, spending source and sextic once;
6. only then apply R-129 and continue through the remaining A13 gates.

R-136 is a scoped T4 exact filtration, replica, raw-telescope, and method boundary result. It proves no production one-use estimate, no positive headroom, no A13 gate, no Nelson estimate, no interacting measure, no removal limit, and no Sector-A closure.

14. Result footer

Result ID:	A13-CLASSII-COMMON-HEAT-REPLICA-RAW-SEQUENTIAL-OWNER-BOUNDARY (R-136).
Claim:	A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement:	Conditional centring contracts after deterministic or retained-measurable operators, but the literal post-heat R-088 atom is retained-field measurable and has zero R-125 future residual. Common-heat future replicas recover the exact conditional variance and owner half. The complete raw owner telescopes pathwise in each replica only when the low endpoint and replica-specific later feedback are retained. Forward Doob/OU coordinates are reused; ambient product PSD is sufficient but not graph-necessary.
Scope:	Fixed finite cutoff and positive floor; one legal retained sigma-field per owner; deterministic R-088 target heat for the zero-residual statement; common heat for the two-replica Hilbert coordinate; complete same-root and exhaustive-output recombination; finite raw sequential paths with feedback recomputed separately in both future replicas.
Dependencies:	R-079, R-084, R-088, R-099, R-102, R-104, R-119, R-123, R-125, R-127, R-128, R-129, R-132, and R-135.
Evidence grade:	ANALYTIC/EXACT/EXECUTED with primary, non-importing independent, and integrated executable checks, a rendered PDF, visual QA, manifest, and release gate.
Reproduction command:	E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/a13_classii_common_heat_replica_raw_sequential_owner_boundary_verify.py
Expected output:	R-136 integrated 225/225 PASS with primary 71/71, non-importing independent 71/71, and aggregate 367/367 pinned by the manifest; PDF and generated-surface gates PASS.

Negative boundary: Post-heat conditional means do not determine raw future variance. The literal post-heat atom cannot be future-centred into the missing R-125 coordinate.

Falsification gate: Any failed projection, commutator, replica factor, raw endpoint telescope, feedback correction, graph compression, authority pin, or scope firewall.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: No production raw spatial intertwiner, one-use q sum, signed forest-current bound, positive headroom, low/matching/gap/anchor data, A13 gate, Nelson theorem, measure theorem, removal limit, or Sector-A closure is proved.

Proof complete: false. Sector A closed: false.

New negative: NG-2026-07-31-A13-POSTHEAT-MEAN-ONLY-FUTURE-VARIANCE-RECOVERY.

Explorations: EXP-000533--EXP-000539.

Next required action: Prove A13-CLASSII-COMMON-HEAT-REPLICA-RAW-CURRENT-SEQUENTIAL-ONE-USE-BOUND, then certify finite-collar headroom and the complete low/gap/anchor data.